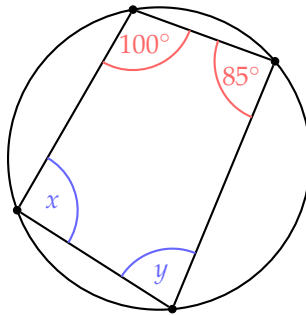


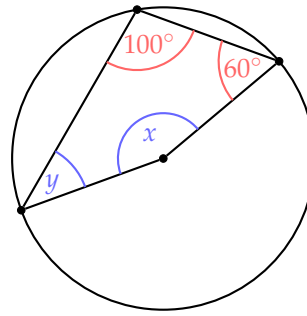
Questions

Q1.) For the circles below, calculate the angles x and y in degrees. State any circle theorems used to find your answers.

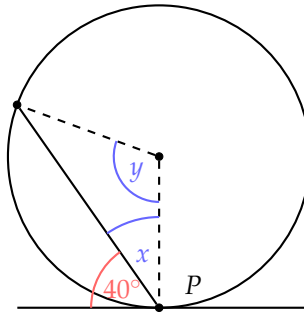
(a)



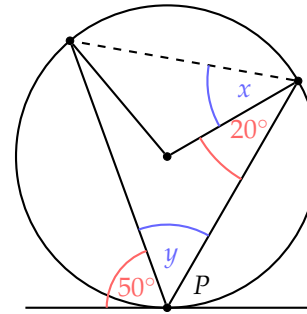
(b)



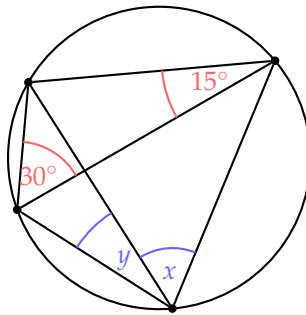
(c)



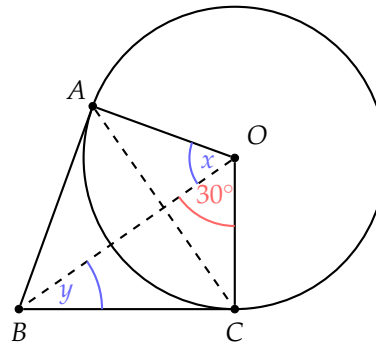
(d)



(e)



(f) The line segments AB and CB are tangent to the circle below.



Q2.) Use circle theorems to explain why the quadrilateral in Q1(b) is not cyclic.

Q3.) Name the type of quadrilateral shown in Q1(f).

Q4.) The radius of the circle in Q1(f) is 5 cm. Using trigonometry, or otherwise, calculate the area of the quadrilateral $OABC$ to 3 significant figures.